HK IMO Training 2021-2022
Problem Set 15
For 22 January 2022

Problem 57. Let p be a prime number. Let $a_1, a_2, \ldots$ be a sequence of positive integers such that

$$a_{n+2} = \frac{a_{n+1}^2 + p}{a_n}$$

for $n = 1, 2, \ldots$. Prove that a_{n+1} divides $a_n + a_{n+2}$ for any positive integer n.

Problem 58. Let A and B be two points on a circle Γ. Consider a variable point C on the major arc of AB on Γ. Let $D \neq C$ be the point on the line BC such that $AC = AD$. Let $E \neq C$ be the point on the line AC such that $BC = BE$. The circle with centre D and passing through C meets the circle with centre E and passing through C again at F. The line passing through C and perpendicular to CF intersects the line AB at P. Let M be the midpoint of AB. The circumcircle of $\triangle CPM$ meets Γ again at Q. Prove that CQ passes through a fixed point independent of C.

Problem 59. Let $S_1, S_2, \ldots, S_{999}$ be subsets of S such that $|S_i \cap S_j| = 1$ for any $i \neq j$. Suppose for every $x \in S$, there are exactly three subsets S_i containing x. Prove that there is a subset T of S such that $|T| = 250$ and $|T \cap S_i| \leqslant 1$ for any i.

Problem 60. Let $n \geqslant 3$ be a positive integer, and let $a_1, a_2, \ldots, a_n, c$ be real numbers such that $a_2 = a_{n-1} = 0$. Prove that

$$\sum_{j=1}^{n-2} |a_j - ca_{j+1} - a_{j+2}| \geqslant |a_1| - |a_n|\,.$$